

# Araz Mehrabani

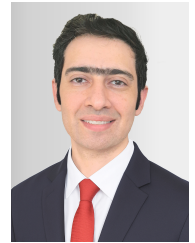
Mechanical Engineer | Wind Energy · Loads & Simulation · Wind Farm Planning

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📅 Date of birth: 28.07.1989



## PROFILE & CORE COMPETENCIES

Mechanical engineer and M.Sc. Wind Energy Engineering candidate combining **wind-turbine loads/aeroelastic simulation** with **wind-farm planning and site assessment**. Hands-on workflows in OpenFAST, windPRO, QGIS and Python from site screening and energy-yield assessment through DLC simulation, component loads and fatigue/extreme evaluation.

## TECHNICAL FOCUS

**Wind Loads & Dynamics:** OpenFAST; IEC 61400-1 / DNV-ST-0437 DLC campaigns; component loads; MLife/MExtremes; rainflow and DEL evaluation.

**Wind Farm Planning:** windPRO; roughness, contours, wind resource, wakes, AEP, noise/shadow, curtailment and layout optimization.

**GIS & Site Screening:** QGIS raster/vector, Shapefile/GeoTIFF, buffer, clip/intersect, protected areas, grid/road access.

**Data & Automation:** Python (NumPy, pandas, Matplotlib, SciPy), CSV/Excel processing, batch simulation workflows, Excel (Pivot/Power Query) and Power BI.

## PROFESSIONAL EXPERIENCE

**Development Engineer** — Sabaniroo, Tehran

01/2022–02/2023

- Set up/adapted **OpenFAST** models and multi-seed DLC campaigns to **DNV-ST-0437**; extracted component channels and evaluated fatigue/extreme loads using MLife/MExtremes, supported by a **Python** workflow for several hundred simulations and approx. **80% less manual effort**.
- Performed **QGIS** site screening in southeastern Iran using raster/vector layers, buffer/clip/intersect and distance analysis for wind potential, protected areas, grid connection, road access and site accessibility.

**Mechanical Engineer** — TehranRigzar, Tehran

03/2017–12/2021

- Built an **ANSYS** line-body/beam-element model of a vibrating screen; evaluated stresses, natural frequencies and mode shapes to avoid resonance, supported by experimental spring testing.

**Working Student – Mechanical Design** — Ario Fidar Maham Co., Tehran

11/2012–05/2013

- Recreated a 40x90 jaw crusher in **CATIA V5**, including individual components, assemblies and manufacturing drawings.

## SELECTED WIND PROJECTS & RESEARCH

**Master's Thesis (ongoing; expected 11/2026) – Rotor as a Sensor** — Python/OpenFAST pipeline using Coleman/Feingold transformation for 0P/1P/2P blade-load harmonics, filter comparison and FFT/order diagnostics; wind inflow (shear, veer and wind direction) estimation.

**20 MW Offshore Wind Turbine – Loads & Dynamics** — Adapted the IEA 15 MW reference to a 20 MW monopile concept; several DLC simulations, component-channel evaluation, MExtremes processing; coordinated load interfaces and technical boundary conditions across development teams.

**Advanced Wind Farm Planning – Hochschule Flensburg** — End-to-end windPRO workflow for wind resource, turbine siting, wakes/AEP, layout optimization, noise/shadow and curtailment assessment.

## EDUCATION

**M.Sc. Wind Energy Engineering**, Hochschule Flensburg — current grade: 1.9

03/2023–11/2026

**M.Sc. Mechanical Engineering**, Hakim Sabzevari University — Control & Robotics

09/2014–01/2017

**B.Sc. Mechanical Engineering**, Islamic Azad University — Structural Dynamics

09/2008–06/2013

## LANGUAGES

German: C1 (actively improving). English: C1 / IELTS 7.0. Persian: Native.